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Introduction

1. Purpose of CTU-Buddy website

CTU-Buddy aims to enhance the university experience by providing a comprehensive educational platform for students, faculty and alumni at City Technical University (CTU) by providing a comprehensive and interactive platform. Its purpose is to facilitate networking, academic collaboration, and resource sharing among its users. The website will centralize essential features such as timetable management, user directory, academic collaboration, resource sharing, discussion forums, and contact information among its users. Designed to support students' academic success and facilitate effective communication within the CTU community, CTU -Buddy exists to foster a connected community among its stakeholders.

2. Project Scope

The project encompasses the design, development, and deployment of the multi-functional educational CTU-Buddy website. It will include:

- Back-end features for coder/administrator:
 - User registration and profile management
 - Directory of users with search and filter options
 - Event calendar and management tools
 - Discussion forums and messaging systems
 - Resource sharing (e.g., study materials, research papers) ○ Administrative backend for managing content and users
- Front-end features for non-administrative user:
 - Home Page: An interactive and immersive central hub with multimedia elements. ○ Timetable Page: An interactive timetable for managing schedules.
 - Discussion Page: A platform for academic discussions and collaborative learning where information could be posted.

- Share Resources Page: A section for uploading and downloading educational materials.
- About Us Page: Information and background about the system and development team.
- Contact Us Page: A contact form and a campus location map for students in need of help or looking to reach out.

The scope includes front-end and back-end development, ensuring responsive design, security, and user-friendly navigation.



3. Team members with Roles and Responsibilities

- **Project Manager:** Oversees the planning of the project, ensures proper execution of project and that deadlines are met, and coordinates between team members and stakeholders.
- **Lead Developer:** Leads the development team, handles complex coding tasks and manages code quality, and ensures technical challenges are solved.
- **Front-End Developer:** Designs and implements the user interface and user experience aspects of the website to ensure visually appealing and responsive designs.
- **Back-End Developer:** Handles server-side logic, manages databases, and integrates front-end features.
- **UI/UX Designer:** Creates wireframes and prototypes to ensure a seamless and smooth user experience across all pages by designing the user interface.
- **Quality Assurance Specialist:** Tests, identifies and fixes bugs in the website, usability issues, and ensures overall quality to ensure smooth website functionality.
- **Content Manager:** Oversees the content creation and management of website content (such as the About Us, Share Resources Pages etc.), including text, images, and multimedia to ensure accurate information.

Project Description

1. User Needs

- **Students:** Need an efficient way to connect with peers, find a platform to share and access academic resources, a safe space for academic discussions, a way to contact support when needed and necessary and a way to stay informed about events and deadlines.
- **Faculty:** Require tools and platforms to manage and share academic resources with students and other faculty, engage with students, provide feedback and organize academic events.
- **Alumni:** To seek opportunities for networking, staying updated with CTU developments, and giving back to the community. And may also benefit from accessing academic resources and participating in discussions.

2. Project Goals

- To create a user-friendly, engaging and interactive platform that addresses the needs of the CTU community. Such as an interactive website that supports students' academic needs.
- To enhance communication and collaboration among students, faculty, and alumni.
- To provide easy access to academic resources and event information by streamlining the management of schedules, resources, and academic discussions.
- To foster a connected and collaborative learning environment at CTU.
- To ensure the website is secure, reliable, and scalable.

3. Project Purpose

The purpose of the CTU-Buddy website is to strengthen and unify the CTU community by providing a centralized platform that supports networking, collaboration, and information sharing. Which is aimed at enhancing the learning experience at CTU by facilitating schedule management, resource sharing, and academic discussions. It aims to improve

the educational experience and foster a supportive and engaged community. The website also aims to improve communication and collaboration among students, faculty, and alumni.

System Features and Requirements

1. Functional Requirements

- Back-End Functional Requirements for coder/Administrator:
 - User Authentication: Users are able to register, log in, and manage their accounts safely and securely.
 - User Directory: A searchable and filterable directory of users with profiles.
 - Event Management: Tools for the creation, management, and RSVPing to events.
 - Discussion Forums: A platform for users to post and participate/reply in discussions.
 - Messaging System: Private, public and group messaging capabilities.
 - Resource Sharing: Upload, download, and share academic materials.
- Front-End Functional Requirements for Non-Administrative user:
 - Home Page: Displays key features, recent updates, and engaging multimedia content.
 - Timetable Page: Allows for students to view and manage their weekly schedules using an aesthetically pleasing table format.
 - Discussion Page: Supports threaded discussions and posts, user profiles, and moderation tools.
 - Share Resources Page: Enables users to upload, categorize, and download educational resources in PDF format.
 - About Us Page: Provides information about the goals and development team of the project.
 - Contact Us Page: Includes a contact form for inquiries and a map showing the CTU campus locations.

2. System Features / Non-Functional Requirements

- Performance: The website handles a high number of concurrent users and performs efficiently under various loads.
- Scalability: The system must be scalable and designed to accommodate and handle future growth in user numbers and content.
- Security: Implements robust security measures for user data and interactions. For example, data encryption, secure authentication, and regular security updates are essential.
- Usability: intuitive and accessible interface, with a responsive design for various devices and users of all levels.

- **Compatibility:** The website must be compatible with major web browsers, various devices and operating systems.
- **Accessibility:** Ensures adherence with accessibility standards and ensures all users, including those with disabilities are supported and are able to use the site effectively.

3. Project Planning

- **Phase 1: Planning (2 weeks):** Describes the project goals and objectives, requirements, and scope. Develops a project plan and schedule deployment of project.
- **Phase 2: Design (4 weeks):** Creates wireframes, prototypes, and finalizes design specifications for each page.
- **Phase 3: Development (8 weeks):** Builds the front-end and back-end of the website and further integrates features.
- **Phase 4: Testing (3 weeks):** Performs a thorough and comprehensive testing, which includes functional, usable, and secure testing. This is meant to identify and resolve existing or potential issues.
- **Phase 5: Deployment (2 weeks):** Deploys the website to a live environment and ensures final checks performance.
- **Phase 6: Maintenance and Support (Ongoing):** Provides ongoing support. Monitors website performance, addresses issues, and makes needed updates to the website.

System Core Component Breakdown Analysis

The **CTU-Buddy Educational System** is an intricately planned online platform with the goal of improving the education of students at City Technical University (CTU). Listed are the main elements and characteristics of the system that align with the project requirements:

1. Home Page

- **Purpose:** Acts as the main hub for the CTU-Buddy platform.
- **Design:** A visually appealing interface with images, videos, and animations to engage users.
- **Features:**
 - Key information about CTU-Buddy, its features, and benefits.
 - Navigation elements and links to other sections of the platform.
 - Footer with easy access to all pages.

2. Timetable Page

- **Purpose:** Allows students to organise their timetable schedules.
- **Design:** Structured timetable designed with HTML tables and CSS for a user-friendly aesthetic.

- **Features:**
 - Shows date, time, modules, and weekdays.
 - Design that is user-friendly and easy to use across mobile phones, tablets, and computers.
 - Helps students keep track of weekly schedules effortlessly.

3. About Us Page

- **Purpose:** Builds connection and trust with the platform users by providing insights into the system and development team.
- **Content:**
 - Overview of CTU-Buddy's purpose, vision, and mission.
 - Team members and their roles in the development of the system.
 - Personal bios to introduce the developers and contributors to users.

4. Contact Us Page

- **Purpose:** Allows users to communicate directly with the team.
- **Features:**
 - Contact form for users to submit inquiries, suggestions, or feedback.
 - Integrated Google Map showcasing the CTU campus location for easy reference.
 - Provides email, phone, and other contact details.

5. Share Resources Page

- **Purpose:** Enables students to upload, share, and access educational resources.
- **Features:**
 - Secure PDF file uploads with categorization for easy organization.
 - Search functionality to help users find specific resources.
 - Download options for offline access to shared resources.

6. Discussion Page

- **Purpose:** Facilitates collaborative learning through discussion forums.
- **Features:**
 - Threaded discussions where students can post and respond to study-related questions.
 - User profiles to create a sense of community.
 - Moderation tools to ensure productive, respectful discussions.

7. Testing and Debugging

- **Goal:** Ensure seamless user experience and functionality across all devices and pages.
- **Process:**

- Comprehensive testing of each feature, including responsiveness, security, and user interactions.
- Debugging issues and refining both visual and functional elements of the platform.

8. Project File Organization and Documentation

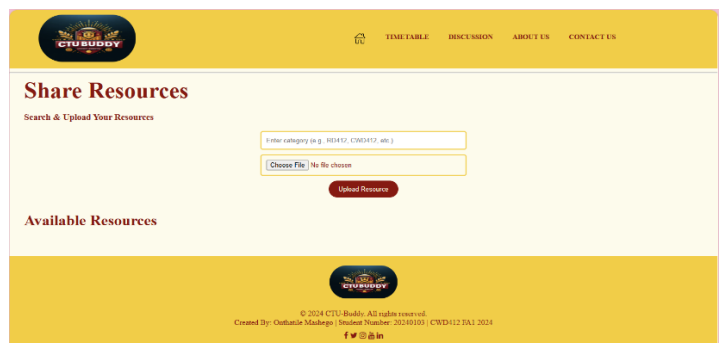
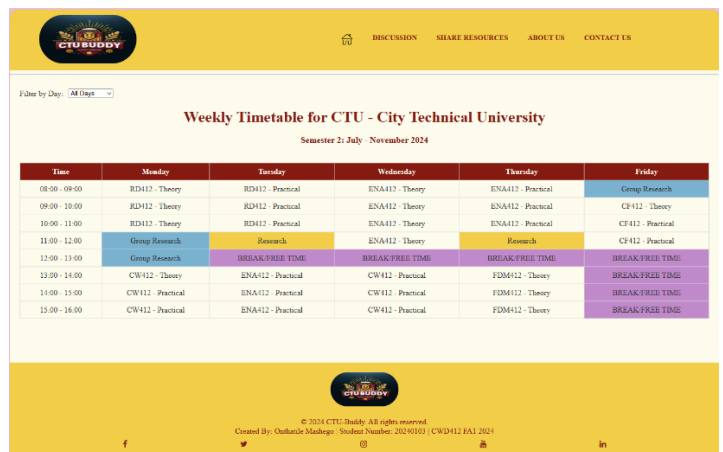
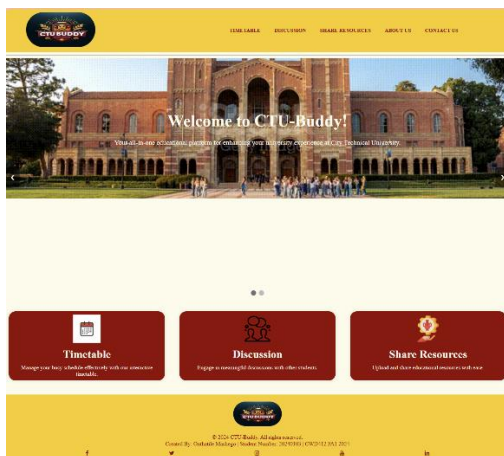
- **Folder Structure:** Includes all website files, team member documentation, and a detailed **SRS (Software Requirements Specification)** document.
- **Submission:** The folder is compressed for easy upload and submission.

9. Impact and Benefits

- **CTU-Buddy** will revolutionize the educational experience by:
 - Providing a **convenient platform** for students to organize their class timetable schedules, participate in conversations and discussion forums, and share resources with other students.
 - Creating a supportive learning environment with interactive features.
 - Encouraging academic collaboration and organization, fostering student success at CTU.

This system, once launched, is expected to greatly improve students' engagement and streamline access to resources, ultimately enhancing the overall academic experience at City Technical University.

Below is an picture extract of some, not all, of the CTU-Buddy website design project explained above:



This outline provides a comprehensive view and reflection of the CTU-Buddy website project, its goals, its scope and its key requirements based on given scenario and specifications. Adjust the timelines and specific details according to your specific needs, constraints and the project's actual progress